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1 #include <stdio.h>
2 int main() {
3     int alloc[10][10], max[10][10], need[10][10];
4     int avail[10], finish[10] = {0};
5     int safeSeq[10];
6     int n, m, i, j, k = 0;
7     printf("Enter Number of Processes: ");
8     scanf("%d", &n);
9     printf("Enter Number of Resources: ");
10    scanf("%d", &m);
11    printf("\nEnter Allocation Matrix:\n");
12    for(i = 0; i < n; i++) {
13        for(j = 0; j < m; j++) {
14            scanf("%d", &alloc[i][j]);
15        }
16    }
17    printf("\nEnter Maximum Matrix:\n");
18    for(i = 0; i < n; i++) {
19        for(j = 0; j < m; j++) {
20            scanf("%d", &max[i][j]);
21        }
22    }
23    printf("\nEnter Available Resources:\n");
24    for(i = 0; i < m; i++) {
25        scanf("%d", &avail[i]);
26    }
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25     scanf("%d", &avail[i]);
26 }
27 for(i = 0; i < n; i++) {
28     for(j = 0; j < m; j++) {
29         need[i][j] = max[i][j] - alloc[i][j];
30     }
31 }
32 int count = 0;
33 while(count < n) {
34     int found = 0;
35     for(i = 0; i < n; i++) {
36         if(finish[i] == 0) {
37             for(j = 0; j < m; j++) {
38                 if(need[i][j] > avail[j])
39                     break;
40             }
41             if(j == m) {
42                 for(k = 0; k < m; k++)
43                     avail[k] += alloc[i][k];
44
45                 safeSeq[count++] = i;
46                 finish[i] = 1;
47                 found = 1;
48             }
49         }
50     }
51     if(found == 0) {
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45         safeSeq[count++] = 1;
46         finish[i] = 1;
47         found = 1;
48     }
49 }
50 }
51 if(found == 0) {
52     printf("System is NOT in Safe State\n");
53     return 0;
54 }
55 }
56 printf("\nSystem is in Safe State\nSafe Sequence: ");
57 for(i = 0; i < n; i++) {
58     printf("P%d ", safeSeq[i]);
59 }
60 return 0;
61 }
```

Enter Number of Processes: 3

Enter Number of Resources: 3

Enter Allocation Matrix:

1 0 4

3 8 9

5 6 7

Enter Maximum Matrix:

4 8 9

5 8 3

0 8 2

Enter Available Resources:

3 3 2

System is in Safe State

Safe Sequence: P1 P2 P0

Process exited after 42.8 seconds with return value 0

Press any key to continue . . . |